JOEL WILLIAMS

St. Joseph's College of Engineering and Technology, Kerala, India (B. Tech in AI & Data Science)

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OVERVIEW

B. Tech candidate in Artificial Intelligence & Data Science who builds AI-powered applications end to end, from machine learning and deep learning models to computer vision systems, reinforcement learning agents, and full-stack software. Fluent in Python, Java, and SQL, with hands-on command of the full ML pipeline, data preprocessing, model training, and deployment into real-world applications. Driven by a genuine curiosity for software engineering and intelligent systems.

SKILLS

Python, Java, SQL, ML & AI application development, DBMS, C, C++, MS Excel, MS Power point

INTERNSHIP

AI & DS Intern – PACELAB Technologies Limited, India June 2025

- Developed AI and data-driven solutions with cross-functional teams, building ML models using Python, Scikit-learn, Pandas, and NumPy.
- Gained hands-on experience in Python, ML workflows, and software development practices, working with real-world datasets such as house price prediction and evaluating models using accuracy (91%), precision, and other metrics.

PROJECTS

DUM-E: the AI Assistant Oct 2025 - Present

- AI-powered desktop assistant (Jarvis-inspired) with voice interaction, app control, web search, and LLM integration, built using SpeechRecognition and Ollama (Llama 3), supporting 15+ voice commands with ~1-3s response time.
- Developed an additional voice- and face-recognition-based password system for user authentication and security.

Call of Duty Game Playing Agent Apr-Oct 2025

Reinforcement learning agent for autonomous gameplay using DQN and PPO-based approaches with a CNN for screen-capture/OpenCV-based game-state interaction; trained over multiple episodes, achieving 86%+ accuracy on headshot execution.

Emotion Recognition & Hand Gesture Recognition System Sep 2025

Real-time computer vision system for facial emotion detection and hand gesture recognition using CNN, the FER2013 dataset, OpenCV, and MediaPipe, achieving ~90% accuracy at 20-30 FPS.

Smart Assistive Home Automation System Oct-Dec 2024

AI-powered eye movement and hand gesture recognition system enabling paralyzed individuals to control home appliances such as lights and fans, using ESP32, relay modules, OpenCV, and MediaPipe, supporting 5+ gestures with real-time control.

Plant Leaf Disease & Cure Detection System Feb-Mar 2024

Deep learning-based plant disease detection with treatment recommendations using MobileNetV2 on the PlantVillage dataset for multi-class disease classification, achieving 88% accuracy.

FitJoe Nov 2023

Fitness tracking web application (FastAPI + MySQL) featuring user authentication, workout logging, calorie tracking, and progress monitoring.

Project Leader – Air Ticket Reservation System Mar 2022 – Feb 2023

- Led team of 4 to design and develop an air ticket reservation system with integrated database functionality using Java, Servlets, and MySQL
- Improved booking efficiency by optimizing workflows and enhancing the overall user experience through database query optimization and streamlined booking workflows.

EDUCATION

Year	Degree/ Certificate	Institution	Marks
2027	B. Tech Artificial Intelligence & Data Science	St. Joseph's College of Engineering & Technology, Kerala, India	7.3/10
2023	Class XII, CBSE	Alphonsa Senior Secondary, Kozhikode, Kerala	87%
2021	Class X, CBSE	Pallotti Hill Public School, Kozhikode, Kerala	92%

EXTRA CURRICULAR ACHIEVEMENTS AND INTERESTS

- **Global Nominee – NASA Space Apps Challenge 2025**: i) Developed AI-based solution for automated exoplanet detection using NASA's open-source astronomical datasets; ii) Trained and evaluated ML models to identify potential exoplanets with improved accuracy.
- **Sports:** i) Athletics: Winner in 100m, 200m competitions across High school; ii) Member- Engineering college Badminton Team